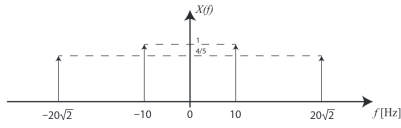


1) a)



Figur 1: Diskret frekvensspektrum.

$$X(f) = \frac{4}{5} \cdot \delta(f - 10 \sqrt{2}) + \delta(f - 0 \pi) + \delta(f + 10 \sqrt{2}) + \frac{1}{5} \delta(f + 20 \sqrt{2})$$

$$b) x(t) = \mathcal{F}^{-1}(X(f)) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df$$

$$\Rightarrow \frac{1}{2\pi} \left(\frac{4}{5} \int_{-\infty}^{\infty} \delta(f - 10 \sqrt{2}) e^{j2\pi f t} df + \int_{-\infty}^{\infty} \delta(f - 10) e^{j2\pi f t} df + \int_{-\infty}^{\infty} \delta(f + 10) e^{j2\pi f t} df + \frac{1}{5} \int_{-\infty}^{\infty} \delta(f + 20 \sqrt{2}) e^{j2\pi f t} df \right)$$

$$\Rightarrow \frac{1}{2\pi} \left(\frac{4}{5} e^{-j40\pi \sqrt{2} t} + \frac{4}{5} e^{j40\pi \sqrt{2} t} + e^{-j20\pi t} + e^{j20\pi t} \right)$$

$$\Rightarrow \frac{1}{2\pi} \left(\frac{4}{5} \cos(40\pi \sqrt{2} t) + \cos(20\pi t) \right)$$

c) Nei sidem det er 2 ulike cos funksjoner.

2)

$$\mathcal{F}\{a x_1(t) + b x_2(t)\}$$

$$= \int_{-\infty}^{\infty} (a x_1(t) + b x_2(t)) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} a x_1(t) e^{-j\omega t} dt + \int_{-\infty}^{\infty} b x_2(t) e^{-j\omega t} dt$$

$$= a \int_{-\infty}^{\infty} x_1(t) e^{-j\omega t} dt + b \int_{-\infty}^{\infty} x_2(t) e^{-j\omega t} dt$$

$$= \underline{\underline{a \mathcal{F}(x_1) + b \mathcal{F}(x_2)}}$$

$$3) \mathcal{L}(h(t)) \Big|_{s=j\omega} = \mathcal{F}(h(t))$$

$$\mathcal{L}(e^{at}) \Big|_{s=j\omega} = \mathcal{F}(e^{at})$$

$$\frac{1}{s-a} \Big|_{s=j\omega} = \int_0^{\infty} e^{at} e^{-j\omega t} dt$$

$$\frac{1}{j\omega-a} = \int_0^{\infty} e^{-(j\omega-a)t} dt$$

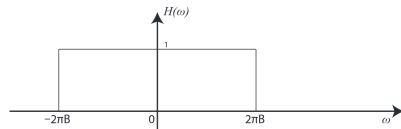
$$= \left[\frac{1}{j\omega-a} e^{-(j\omega-a)t} \right]_0^{\infty}$$

$$= \frac{1}{j\omega-a}$$

Gjelder ikke om systemet ikke er kausalt fordi da vil vi ta fourier fra -uendelig til uendelig og ikke fra 0 til uendelig som gjør at det ikke blir likt.

4)

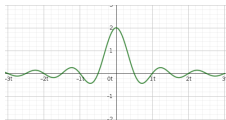
Figur 2 viser frekvensresponsen $H(\omega)$ til et ideelt lavpassfilter med båndbredde B .



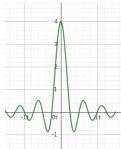
Figur 2: Frekvensspektrum for et ideelt lavpassfilter.

$$\begin{aligned} a) \quad \mathcal{F}^{-1}(H(\omega)) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\omega) e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-2\pi B}^{2\pi B} e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \left(\frac{1}{jt} e^{j\omega t} \Big|_{-2\pi B}^{2\pi B} \right) \\ &= \frac{1}{j2\pi t} \left(e^{j2\pi B t} - e^{-j2\pi B t} \right) \\ &= \frac{\sin(2\pi B t)}{\pi t} \Big|_{\frac{2B}{2B}} \\ &= \underline{\underline{2B \operatorname{sinc}(2Bt)}} \end{aligned}$$

b)



$B=1$

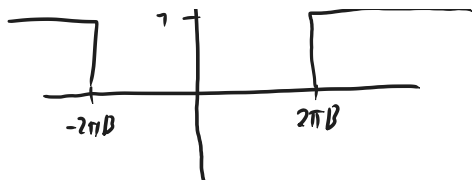


$B=2$

c) Ikke kausisk

5)

$H(\omega)$
|



$$H_{HP}(\omega) = 1 - H_{LP}(\omega)$$

$$\begin{aligned} \mathcal{F}^{-1}(H_{HP}(\omega)) &= \mathcal{F}^{-1}(1) - \mathcal{F}^{-1}(H_{LP}(\omega)) \\ &= \underline{\underline{\delta(t) - h_{LP}(t)}} \end{aligned}$$

7)

$$a) 20 \log\left(\frac{0,5}{1}\right) = \underline{\underline{-6 \text{ dB}}}$$

b)

$$20 \log\left(\frac{|X(f_n)|}{|X(f_0)|} \frac{|H_D(f_n)|}{|H_D(f_0)|}\right) =$$

$$\Rightarrow 20 \log\left(\frac{\sqrt{2}}{2\sqrt{1 + \left(\frac{f_n}{f_0}\right)^{2n}}}\right)$$

$$\Rightarrow n=2 : \underline{\underline{-18,3 \text{ dB}}}$$

$$\Rightarrow n=3 : \underline{\underline{-27,1 \text{ dB}}}$$

Figure 3 viser et symmetrisk signal $x(t)$ i tidsdomænet, som er givet ved

$$x(t) = \begin{cases} 1 - \frac{|t|}{T} & ; |t| \leq T \\ 0 & ; |t| > T \end{cases}$$

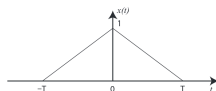


Figure 3: Symmetrisk signal.

Vis at spektret, altså Fouriertransformen, til signalet $x(t)$ er givet ved

$$X(\omega) = T \text{sinc}^2\left(\frac{T\omega}{2\pi}\right).$$

$$T \text{sinc}^2(x) = 2 \sin^2\left(\frac{x}{2}\right).$$

$$\begin{aligned} \mathcal{F}(x(t)) &= \int_{-T}^0 \left(1 + \frac{t}{T}\right) e^{-j\omega t} dt + \int_0^T \left(1 - \frac{t}{T}\right) e^{-j\omega t} dt \\ &= \int_{-T}^0 e^{-j\omega t} dt + \int_{-T}^0 \frac{t}{T} e^{-j\omega t} dt \\ &\quad + \int_0^T e^{-j\omega t} dt - \int_0^T \frac{t}{T} e^{-j\omega t} dt \end{aligned}$$

$$\begin{aligned}
&\Rightarrow \left[-\frac{1}{j\omega} e^{-j\omega t} \right]_{-T}^0 + \left[-\frac{1}{j\omega} e^{-j\omega t} \right]_0^T \\
&\quad + \left[\frac{1}{j\omega} \frac{t}{T} e^{-j\omega t} \right]_{-T}^0 - \frac{1}{T} \int_{-T}^0 e^{-j\omega t} dt \\
&\quad - \left[\frac{1}{j\omega} + \frac{t}{T} e^{-j\omega t} \right]_0^T + \frac{1}{T} \int_0^T e^{-j\omega t} dt \\
&\Rightarrow -\frac{1}{j\omega} e^{j\omega T} + \frac{1}{j\omega} - \frac{1}{j\omega} + \frac{1}{j\omega} e^{-j\omega T} \\
&\quad - \frac{1}{j\omega} e^{j\omega T} + \frac{1}{T} \left[\frac{1}{j\omega} e^{-j\omega t} \right]_{-T}^0 \\
&\quad + \frac{1}{j\omega} e^{-j\omega T} + \frac{1}{T} \left[-\frac{1}{j\omega} e^{-j\omega t} \right]_0^T \\
&\Rightarrow \frac{-2}{j\omega} (e^{j\omega T} - e^{-j\omega T}) + \frac{1}{T} \left(\frac{1}{j\omega} e^{j\omega T} - \frac{1}{j\omega} \right) \\
&\quad + \frac{1}{T} \left(\frac{1}{j\omega} - \frac{1}{j\omega} e^{-j\omega T} \right) \\
&\Rightarrow \frac{1}{j\omega T} (e^{j\omega T} - e^{-j\omega T}) - \frac{2}{j\omega} (e^{j\omega T} - e^{-j\omega T}) \\
&\Rightarrow \left(\frac{1}{j\omega T} - \frac{2}{j\omega} \right) (e^{j\omega T} - e^{-j\omega T}) \\
&\Rightarrow \left(\frac{2(1-2T)}{T} \right) \sin(\omega T)
\end{aligned}$$

Fantasi di magi sa' blir deh
hok nge sine til sluk